



Department of Information Technology

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: II Semester B.Tech IT

Course Code & Title: CS3251 & Programming in C

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Looping Statement

- **Course Outcome:** CO1

- **Topic Learning Outcome:** TLO3

- **Activity Chosen:** Code Debugging

- **Justification:**

Looping is a fundamental concept in programming languages like C. Loops are used to repeat a block of code until the specified condition is met. A loop statement allows programmers to execute a statement or group of statements multiple times without repetition of code. It is necessary to memorize the syntax of various looping statement in C programming. To make the students familiarize with the concept of looping simple C snippet with error is provided to the students. The role of the student is to find the error in the given code.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- For this activity, students were asked to form a team of two members. Different set of questions were designed.
- Each team was given with a C code snippet involving the various kinds of looping statements in C in which there is error in the code along with the desired output is given. The students have to debug the code to remove the errors from them for getting the desired output.
- The sample questions and the student's answers are shown in Fig 1.
- The activity was conducted during the last 15 minutes of the class. In the first 10 minutes the students have to think and identify the missing code and debug it to get the desired output.
- At the end of the activity, the answers for the questions were discussed by the student team members.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	1	2	1	1	3	1	1

(1 – Low

2 – Moderate

3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1	PSO2	PSO3
	3	2	1	1	1	3	1	1
Justification for correlation	The basic knowledge about looping statement is needed.	The students analyze the given partial code.	The students debug the code to answer for the given coding questions.	The students coordinate and demonstrate the activity as a team.	The students communicate the solutions to the others	The concept is used in solving data science problems	The concept is used in IOT, Mobile application.	The concept is used in developing software projects.

- Images / Screenshot of the practice:

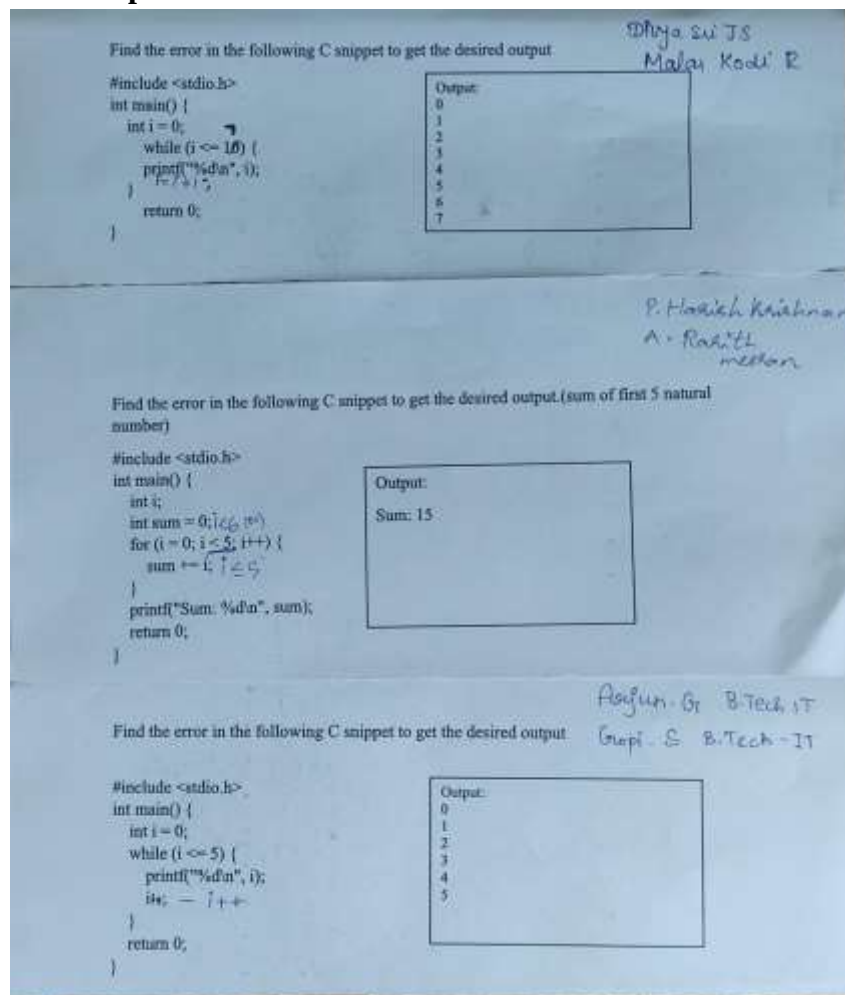


Fig 1: Students Response sheet for Code Debugging

(Divya Sri.J.S, Malarkodi. K, Gopi. S, Arjun.G, A. Rasith meeran, P. Harish Krishnan)

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The students feel it is very interesting to debug the given code. They can easily identify the syntax errors they will frequently encounter during coding.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- It helps the students to remember the syntax and the working of looping statements.

- ***Challenges faced in implementation:***

- Some of the students those who does not understand the looping statement felt difficult to find answers.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/1.GSP_Code-Debugging.pdf
- <https://code.org/curriculum/course1/5/Teacher>
- <https://www.canadalearningcode.ca/lessons/speaking-code-debugging-lesson-6>

Signature of Faculty Member

HOD